

Lecture 20

Strings and the thermodynamic Bethe ansatz

Objectives. *Lecture 19* put one state — the ground state — into the thermodynamic limit. At any temperature above zero there are exponentially many states to choose from, and thermodynamics is the business of counting them. Counting Bethe states is unusually easy, because *Lecture 19* already found the slots: a state is a choice of which vacancies are filled. The count gives an entropy, minimising the free energy against it gives one nonlinear integral equation, and the solution of that equation is the exact equation of state of an interacting quantum gas — exact at every temperature and coupling, and measured on an atom chip. The chain costs more, because its roots are not all real: they form strings, and the one equation becomes infinitely many, which collapse again into the Y-system.

20.1 Counting

Everything in this lecture rests on one sentence of *Lecture 19*: a Bethe state is not a list of rapidities but a choice of *quantum numbers* from the list of *vacancies*. Choose a subset, solve (19.1), and the roots come out. So counting states is counting subsets — and that is a problem with an answer. For the chain the list is finite, of length $N - M$, and *on the branch where ζ increases* the roots occupy M distinct members of it (*Proposition 19.1*, whose second sentence *Remark 19.2* showed to be a hypothesis); for the gas the list is all of $\mathbb{Z} + \frac{N-1}{2}$ and is not finite at all. What the count below uses is neither, but the local *density* of vacancies ρ_t , which both models have.

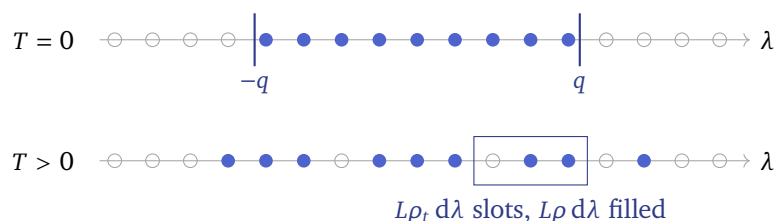


Figure 20.1: A Bethe state is a choice of which vacancies carry roots. At $T = 0$ the choice is forced — a solid block: no holes inside it, no roots outside (*Lecture 19*). At $T > 0$ any subset is allowed, and a cell of the rapidity line with $L\rho_t d\lambda$ slots and $L\rho d\lambda$ of them filled can be filled in $\binom{L\rho_t d\lambda}{L\rho d\lambda}$ ways. The logarithm of the product over cells is the Yang–Yang entropy.

Fix a coarse-graining: cut the rapidity line into cells of width $d\lambda$, small on the scale over which ρ varies but wide enough to hold many vacancies. In the cell at λ there are $L\rho_t(\lambda)d\lambda$

slots and $L\rho(\lambda)d\lambda$ of them carry roots, where L is the number of sites N or the length of the box, as in (19.5). The cells are independent, so

Proposition 20.1 (The Yang–Yang entropy). *The number of Bethe states compatible with the densities ρ, ρ_h is $\prod_{\text{cells}} \binom{L\rho_t d\lambda}{L\rho d\lambda}$, and its logarithm is $Ls + o(L)$ with*

$$s = \int [\rho_t \ln \rho_t - \rho \ln \rho - \rho_h \ln \rho_h] d\lambda = \int \rho_t [-\vartheta \ln \vartheta - (1 - \vartheta) \ln(1 - \vartheta)] d\lambda, \quad (20.1)$$

the entropy per site (or per unit length), with $\vartheta = \rho/\rho_t$ the filling fraction of Definition 19.8.

· *proved as a count, from the string and completeness hypotheses named below*

Proof (★). Stirling gives $\ln \binom{A}{B} = A \ln A - B \ln B - (A - B) \ln(A - B) + O(\ln A)$, so three limits are needed and their order is not free. Cut the rapidity line at $|\lambda| \leq \Lambda$ and take finitely many cells there. Let $L \rightarrow \infty$: the leading terms are $Ld\lambda$ times the bracket in (20.1) and the errors are $O(\ln L)$ in each of finitely many cells, hence $o(L)$. Then refine the cells, which turns the sum into the integral. Then let $\Lambda \rightarrow \infty$, which converges because the entropy density is integrable. Taking the first two in the other order would not do — with one cell per vacancy the errors are $O(L \ln L)$ — and without the cutoff the first step never starts, since the gas’s outermost cell holds infinitely many slots. The second form is the first with $\rho = \vartheta\rho_t$, $\rho_h = (1 - \vartheta)\rho_t$. ■

Three things are worth saying aloud before this is used.

Remark 20.2 (What the count assumes). It assumes, first — the step doing the work — that every way of filling the slots is admissible, with no further extensive restriction within a cell or between cells. A rule against occupying *adjacent* slots, say, would replace $\binom{A}{B}$ by $\binom{A-B+1}{B}$ and change the entropy at leading order. Second, that distinct admissible choices give distinct states and exhaust the spectrum — completeness, which Lecture 15 could only cite. Third, that all choices with the same coarse-grained densities are equally numerous. And fourth, for the chain, the string hypothesis of § 20.5, since without it the list of quantum numbers is not known. None of the four is proved here. What is proved is everything downstream of (20.1).

Remark 20.3 (Two familiar limits of one formula). $\vartheta = 0$ and $\vartheta = 1$ both give zero: an empty region and a full one each admit one arrangement. The maximum is $\ln 2$ per slot at $\vartheta = \frac{1}{2}$. And (20.1) is the entropy of a set of *fermionic* modes — one slot, one particle — even though the gas of Lecture 16 is made of bosons. The exclusion is not in the statistics of the particles but in the Bethe equations, which forbid two equal quantum numbers.

20.2 Minimising the free energy

At temperature T and chemical potential μ the equilibrium state minimises the grand free energy per unit length,

$$g[\rho] = \int (\varepsilon_0(\lambda) - \mu) \rho(\lambda) d\lambda - Ts, \quad (20.2)$$

where ε_0 is the bare energy of one root — $\varepsilon_0 = k^2$ for the gas of Lecture 16, $\varepsilon_0 = -\pi J a_1$ for the chain (§ 20.5). Here μ is the *chemical potential*; no spectral parameter appears in this lecture, and the clash named in Lecture 19 is again harmless.

Remark 20.4 (ρ_h is not yours to vary). Equation (20.2) is a functional of ρ alone. The hole density is fixed by (19.5),

$$\rho + \rho_h = \frac{p'_0}{2\pi} + \frac{1}{2\pi} \int K(\lambda - \lambda') \rho(\lambda') d\lambda', \quad \text{so} \quad \delta\rho_h = -\delta\rho + \frac{1}{2\pi} \int K \delta\rho. \quad (20.3)$$

The kernel term is the whole of the interaction. Drop it and the variation returns free fermions with the bare dispersion; treat ρ_h as an independent function and the answer is a Boltzmann factor with no exclusion at all. Both are checked in the script, and the second is the error the section is named after.

Theorem 20.5 (Yang–Yang). Write $\varepsilon = T \ln(\rho_h/\rho)$, the pseudo-energy. Then ρ is stationary for (20.2) subject to (20.3) if and only if

$$\varepsilon(\lambda) = \varepsilon_0(\lambda) - \mu - \frac{T}{2\pi} \int K(\lambda - \lambda') \ln(1 + e^{-\varepsilon(\lambda')/T}) d\lambda', \quad (20.4)$$

and then the filling fraction is a Fermi function of the pseudo-energy, and the minimum value of g is

$$\vartheta = \frac{1}{1 + e^{\varepsilon/T}}, \quad \min g = -\frac{T}{2\pi} \int p'_0(\lambda) \ln(1 + e^{-\varepsilon(\lambda)/T}) d\lambda. \quad (20.5)$$

For the gas $p'_0 = 1$ and $-\min g$ is the pressure P .

· proved from [Proposition 20.1](#); that the minimiser gives the true free energy is Yang and Yang's saddle-point argument, in the reading

Proof. From (20.1), $\delta s = \int [\ln(\rho_t/\rho) \delta\rho + \ln(\rho_t/\rho_h) \delta\rho_h]$, and $\rho_h/\rho = e^{\varepsilon/T}$ turns the two logarithms into $\ln(1 + e^{\varepsilon/T})$ and $\ln(1 + e^{-\varepsilon/T})$. Substituting $\delta\rho_h$ from (20.3) and using that K is even,

$$\frac{\delta g}{\delta\rho(\lambda)} = (\varepsilon_0 - \mu) - T \left[\ln(1 + e^{\varepsilon/T}) - \ln(1 + e^{-\varepsilon/T}) \right] - \frac{T}{2\pi} \int K(\lambda - \lambda') \ln(1 + e^{-\varepsilon(\lambda')/T}) d\lambda'.$$

The bracket is ε/T identically, so the vanishing of the derivative is (20.4). The first equation of (20.5) is the definition of ε rearranged. For the second, write $Ts = \int \rho\varepsilon + T \int \rho_t \ln(1 + e^{-\varepsilon/T})$, use (20.4) to replace $\varepsilon_0 - \mu$ in (20.2), and use (20.3) to replace ρ_t : the two double integrals are equal by the symmetry of K and cancel, leaving $g = -\frac{T}{2\pi} \int p'_0 \ln(1 + e^{-\varepsilon/T})$. The weight p'_0 is not decoration — it is what § 20.5 needs, and dropping it would make the chain's free energy diverge. Finally the stationary point is a minimum: writing $\sigma(\vartheta) = -\vartheta \ln \vartheta - (1 - \vartheta) \ln(1 - \vartheta)$, the integrand of (20.1) is $\rho_t \sigma(\rho/\rho_t)$, the perspective of the concave σ and therefore jointly concave in (ρ, ρ_t) ; and $\rho \mapsto (\rho, \rho_t[\rho])$ is affine by (20.3). So s is concave in ρ , g is convex, and the admissible set $\{0 \leq \rho \leq \rho_t[\rho]\}$ is convex. Convexity gives minimality, not uniqueness — the perspective is not strictly concave, its second variation $-(\rho_h \delta\rho - \rho \delta\rho_h)^2 / \rho \rho_h \rho_t$ vanishing whenever $\delta\rho = \vartheta \frac{1}{2\pi} K \delta\rho$. That (20.4) has exactly one solution at every $c > 0$, $T > 0$ is Yang and Yang's, and is cited. ■

Equation (20.4) is nonlinear — the unknown sits inside a logarithm inside the integral — and that is the price of the temperature. Everything else about it is a gift: one scalar equation, one scalar unknown, and no reference to N or to any particular state. And what the second half of (20.5) says is that the minimisation is a Legendre transform. The minimum value of

$g = e - \mu n - Ts$ is the grand potential density, $-P(\mu, T)$; so P is the Legendre transform of the energy density in both n and s , its partners come back by differentiating,

$$n = \frac{\partial P}{\partial \mu}, \quad s = \frac{\partial P}{\partial T}, \quad \frac{E}{L} = Ts + \mu n - P, \quad (20.6)$$

and one scalar function of two variables carries the whole thermodynamics. Students lose the last of these by trying to compute E/L from $\int \varepsilon \rho$, which is not it: ε is not an energy.

Corollary 20.6 (Dressing at finite temperature). $\varepsilon' = (\varepsilon'_0)^{\text{dr}}$ at every temperature, with ϑ of (20.5) in (19.13), and $2\pi\rho_t = (p'_0)^{\text{dr}}$. The effective velocity at temperature T is defined, for the gas, by the same ratio as at $T = 0$,

$$v^{\text{eff}}(k) = \frac{\varepsilon'(k)}{2\pi\rho_t(k)}, \quad (20.7)$$

which reduces to (19.14) when ϑ is an indicator. For the chain the denominator must be the excitation's own bare momentum dressed, not the counting momentum's (Remark 19.10).

Proof. Differentiate (20.4) in λ , move the derivative onto μ' by $\partial_\lambda K(\lambda - \lambda') = -\partial_{\lambda'} K(\lambda - \lambda')$ and integrate by parts; the boundary terms vanish because $K \rightarrow 0$. What is left involves $\partial_{\lambda'} \ln(1 + e^{-\varepsilon'/T}) = -\vartheta \varepsilon'/T$, and the T cancels:

$$\varepsilon'(\lambda) = \varepsilon'_0(\lambda) + \frac{1}{2\pi} \int K(\lambda - \lambda') \vartheta(\lambda') \varepsilon'(\lambda') d\lambda',$$

which is (19.13) for ε'_0 . The second statement is Proposition 19.9, whose first half holds for any ϑ ; the third is a definition. *That the ratio is a propagation speed is not proved here.* There are excitations at $T > 0$ — moving a root from λ_h to λ_p costs $\varepsilon(\lambda_p) - \varepsilon(\lambda_h)$, backflow included — cited: Korepin, Bogoliubov and Izergin §1.8 — but their transport interpretation is Lecture 23's, not this lecture's. ■

Remark 20.7 (The same identity, a different proof). Proposition 19.9 got $\varepsilon' = (\varepsilon'_0)^{\text{dr}}$ out of the linear equation (19.13), and needed two hypotheses to do it: $T = 0$, so that ϑ is an indicator, and $\varepsilon(\pm q) = 0$, to kill the boundary terms. Neither survives here — at $T > 0$ the filling fraction is smooth, and the zeros of ε have moved off $\pm q$ (they are at 1.5002, 1.5184 and 1.7133 for the three temperatures of Figure 20.2, against $q = 1.5$). The identity survives because the nonlinear equation supplies ϑ from its own derivative. Quoting Proposition 19.9 at finite temperature would be wrong; deriving it again from (20.4) is right, and the bulk term Lecture 19 warned about is genuinely present and genuinely cancels.

20.3 Two checks

A new equation earns trust by reproducing the two answers already known.

Proposition 20.8 (Impenetrable, and cold). For the gas of Lecture 16, with $\varepsilon_0 = k^2$ and $K = 2c/(k^2 + c^2)$:

1. as $c \rightarrow \infty$, $\varepsilon(k) = k^2 - \mu - 2P/c + O(c^{-2})$ uniformly on compact sets of k , and P tends to the free-fermion pressure $\frac{T}{2\pi} \int \ln(1 + e^{-(k^2 - \mu)/T}) dk$, the correction being $O(1/c)$;
2. as $T \rightarrow 0$ at $\mu > 0$, (20.4) becomes (19.18) and $P \rightarrow -\frac{1}{2\pi} \int_{-q}^q \varepsilon = \mu n - E/L$, with the edge condition $\varepsilon(\pm q) = 0$ deduced, not imposed. (At $\mu < 0$ the limit is the vacuum:

$$\varepsilon = k^2 - \mu > 0 \text{ everywhere, no sea, no zero, } n = P = 0.)$$

Proof (★). Both parts need one imported fact: on any bounded set of parameters there is a C with $\varepsilon(k) \geq k^2 - C$, uniformly · *cited: Yang and Yang (1969)*. It is what makes $\ln(1 + e^{-\varepsilon/T})$ integrable with bounded zeroth and second moments, and both proofs use exactly that.

(1) $\int K dk = 2\pi$ for every c , and $K(k) = \frac{2}{c} + O(c^{-3}k^2)$. With the moments bounded uniformly in $c \geq c_0$, for k in a compact set the integral in (20.4) is $\frac{T}{2\pi} \cdot \frac{2}{c} \int \ln(1 + e^{-\varepsilon/T}) + O(c^{-3}) = 2P/c + O(c^{-3})$; expanding P itself then leaves $O(c^{-2})$. That is a constant shift of μ . (2) $T \ln(1 + e^{-\varepsilon/T}) \rightarrow \max(0, -\varepsilon)$ pointwise and to within $T \ln 2$ uniformly — enough on a compact set, and the tails are handled by the same quadratic bound, *not* by the uniform bound, whose integral over $\mathbb{R}$ is infinite. The limit is supported where $\varepsilon < 0$; since ε is continuous and strictly increasing in $|k|$ · *cited: Yang and Yang (1969)*, that set is an interval, empty at $\mu < 0$ — the vacuum — and otherwise $(-q, q)$ with $\varepsilon(\pm q) = 0$. On it the limit of (20.4) is $\varepsilon = k^2 - \mu + \frac{1}{2\pi} \int_{-q}^q K \varepsilon$, which is (19.18). For P , split (20.5) the same way; $\mu n - E/L$ then follows from $P = Ts + \mu n - E/L$ and $s \rightarrow 0$. ■

Part (1) carries a sentence of physics, and it is the opposite of the one that suggests itself. The shift *raises* μ , so the gas has *more* pressure than the impenetrable one, while hard rods of length a give $P = P_F(\mu - aP)$ and less: matching, $a = -2/c$. At large finite c the gas is free fermions in a box *lengthened* by $2/c$ per particle. The Bethe equations say so directly — $\theta(k) \approx 2k/c$ turns (16.11) into $k_j(L + 2N/c) = 2\pi I_j$ in the rest frame, $\sum_l k_l = 0$ — and so does Lecture 19's $E/N = n^2 e = \frac{\pi^2 n^2}{3} (1 - 4/\gamma + \dots)$, which to this order is $\pi^2 n_{\text{eff}}^2/3$ with $n_{\text{eff}} = n/(1 + 2n/c)$. At $c = 100$, $T = \mu = 1$ the shift accounts for the whole of the 0.6% by which the interacting pressure exceeds the free-fermion one.

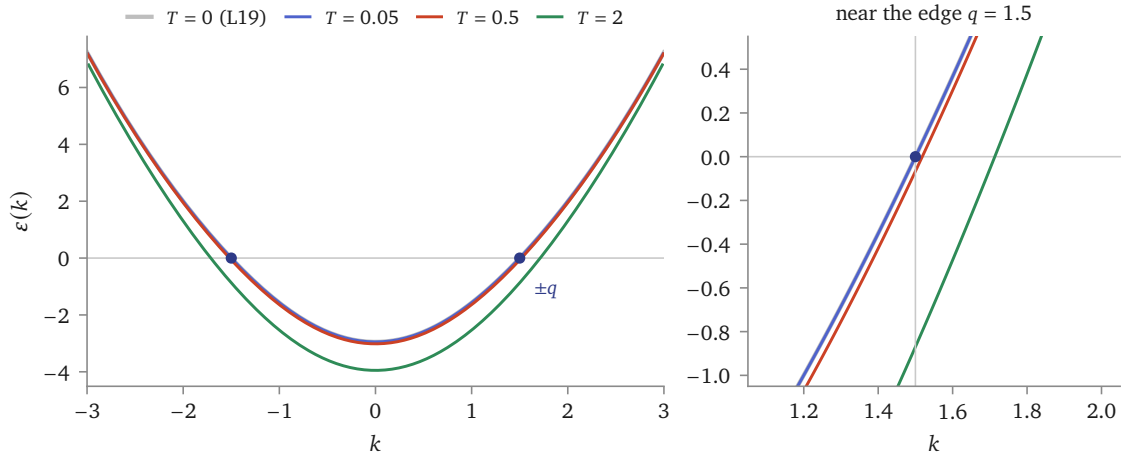


Figure 20.2: The pseudo-energy $\varepsilon(k)$ of (20.4) at three temperatures, $c = 1$ and μ held at the value Lecture 19 gives for $q = 1.5$. Grey: the $T = 0$ dressed energy (19.18), whose zeros are the Fermi edges $\pm q$. At $T > 0$ there are no edges: the curve slides down and $\varepsilon(\pm q)$ moves away from zero, which is why Proposition 19.9 cannot be quoted here.

20.4 Solving it

Equation (20.4) has no closed-form solution and does not need one: it is a fixed-point equation, and the obvious iteration works. It also has one fewer parameter than it looks:

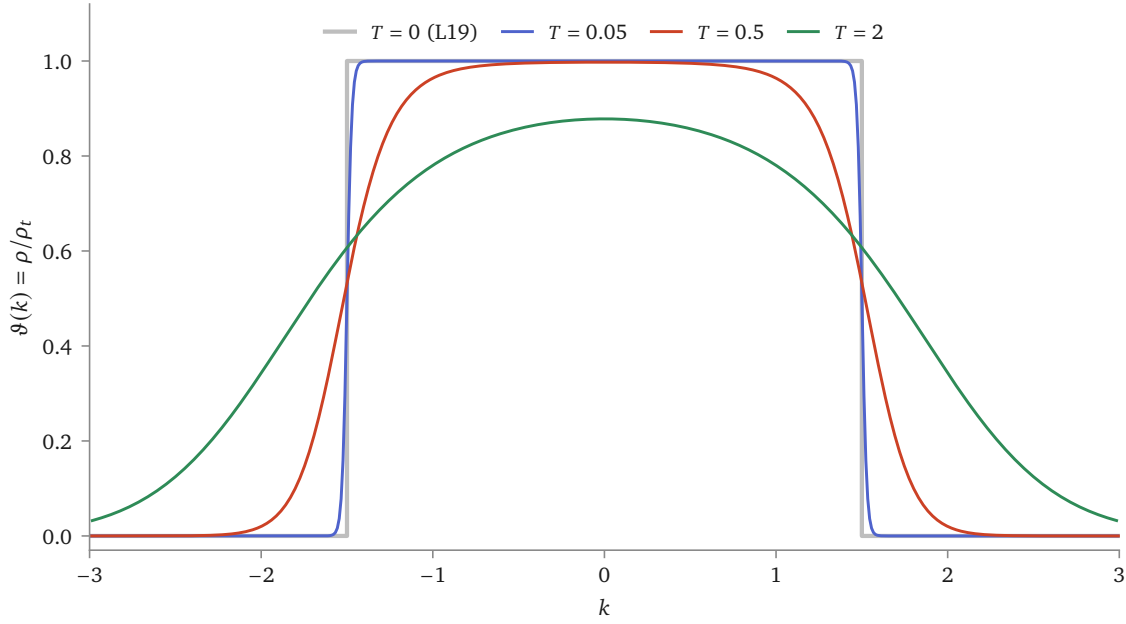


Figure 20.3: The filling fraction $\vartheta = \rho/\rho_t$ for the same three temperatures, against the step function of the $T = 0$ sea. On this axis the crossing region has width $\sim T/|\epsilon'|$ — it widens by a factor ten from $T = 0.05$ to $T = 0.5$ — which is T/v^{eff} only after converting rapidity to dressed momentum, $p' = 2\pi\rho_t$. By $T = 2$ the sea is no longer degenerate at all: $\vartheta < 0.88$ everywhere. ϑ is the one object that carries the state, at $T = 0$ (Lecture 19), here, and out of equilibrium (Lecture 23).

substituting $k = \sqrt{T}\kappa$ and $c = \sqrt{T}\hat{c}$ makes every $\sqrt{T}$ cancel between K and dk' , so that ϵ/T and $P/T^{3/2}$ depend on $(\mu/T, \hat{c})$ alone. One plot is the whole equation of state.

Proposition 20.9 (The iteration contracts). *Let $\epsilon \mapsto \mathcal{F}\epsilon$ be the right side of (20.4). If $\vartheta \leq \bar{\vartheta}$ along the iteration then $\|\mathcal{F}\epsilon - \mathcal{F}\tilde{\epsilon}\|_\infty \leq \bar{\vartheta} \|\epsilon - \tilde{\epsilon}\|_\infty$.*

Proof (★). $\partial_\epsilon \ln(1 + e^{-\epsilon/T}) = -\vartheta/T$, and $\frac{1}{2\pi} \int |K| = 1$ in both models — the one place in this lecture where the sign of K (Remark 19.4) does not matter. So the T cancels and the factor is $\sup \vartheta$. ■

The linearisation being differentiated here is nothing other than the dressing operator $\frac{1}{2\pi}K\vartheta$ of (19.13), so the equation that answers every question about the state is also the one that says how fast the solver converges. The sup-norm bound is crude — what the iteration feels is that operator's spectral radius — and at low temperature it is useless, since ϑ reaches 1. The spectral radius does not: starting from $\epsilon = k^2 - \mu$ at $c = 1$, $\mu = 1$, the residual falls by 0.448 a step at $T = 1$ and by 0.497 at $T = 0.05$, and those are the two spectral radii to three figures, although $\sup \vartheta$ is 0.887 and 1.000. Either way 10^{-14} takes forty-odd steps.

Two features of Figure 20.4 deserve naming. On the dilute side every curve collapses onto $P/T^{3/2} = e^{\mu/T}/2\sqrt{\pi}$: a gas dilute enough is classical, and the coupling cannot be seen. On the degenerate side the curves fan out, and the free-fermion curve is the floor: since $K > 0$, (20.4) gives $\epsilon \leq k^2 - \mu$ pointwise and hence $P \geq P_F$ at the same μ and T . The impenetrable gas has the least pressure there is, which is the statement of Proposition 20.8(1) seen from the other side.

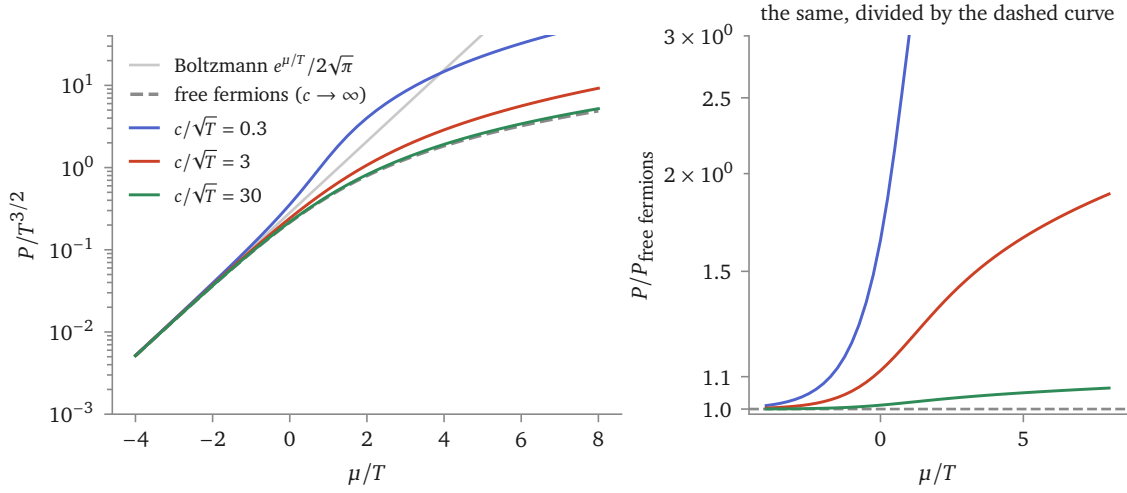


Figure 20.4: The equation of state. By scaling, (20.4) depends on $(\mu/T, c/\sqrt{T})$ only, so one plot of $P/T^{3/2}$ against μ/T carries every temperature. Dashed: the impenetrable limit $c \rightarrow \infty$, free fermions. Faint: the Boltzmann asymptote $e^{\mu/T}/2\sqrt{\pi}$, which every curve joins on the dilute side, interactions or not. The approach to the dashed curve is governed by $\gamma = c/n$, not by $c/\sqrt{T}$: at $\mu/T = 8$ the gas is degenerate and $c/\sqrt{T} = 30$ is still 7% above it.

The experiment

Equation (20.4) is not a model equation of state; it is the equation of state of a real gas. Ultracold ^{87}Rb on an atom chip, confined so tightly in two directions that only the lowest transverse mode is populated, is the Lieb–Liniger gas of Lecture 16 with c set by the transverse confinement. Measuring the *in situ* density profile of such a cloud, and reading the trap’s slowly varying potential as a local chemical potential $\mu(x) = \mu_0 - V(x)$, turns one image into a curve of n against μ at fixed T . Van Amerongen and co-workers did this in 2008 for a weakly interacting gas on an atom chip, fitting only μ_0 and T : no interaction parameter is free, since c follows from the transverse confinement. One complication is instructive — at their temperatures the higher transverse modes are not empty, so what is fitted is $n_{\text{YY}}(\mu(x), T) + \sum_{j \geq 1} (j+1)n_{\text{ideal}}(\mu(x) - j\hbar\omega_{\perp}, T)$: Yang–Yang for the one-dimensional gas, an ideal gas for everything above it. With that, the profiles are reproduced across the crossover from the quasi-condensate to the nearly ideal gas — where mean-field theory has nothing to say · cited: van Amerongen et al., *Phys. Rev. Lett.* **100** (2008) 090402.

20.5 Strings

Everything so far used only real roots, and for the gas that is a theorem (Proposition 16.4). The chain is different. Lecture 15 already met a solution with $\lambda_{1,2} = x \pm \frac{i}{2} + O(\delta)$ and δ exponentially small in N (15.21), and its argument — an imaginary part forces a nearly vanishing denominator — applies to any number of roots.

Definition 20.10 (Strings, and the hypothesis). An n -string is a set of n roots $\lambda_{\alpha}^n + \frac{i}{2}(n+1-2a)$, $a = 1, \dots, n$, with real centre λ_{α}^n . The *string hypothesis* is that as $N \rightarrow \infty$ the roots of the solutions that matter thermodynamically organise into strings, up to deviations exponentially small in N .

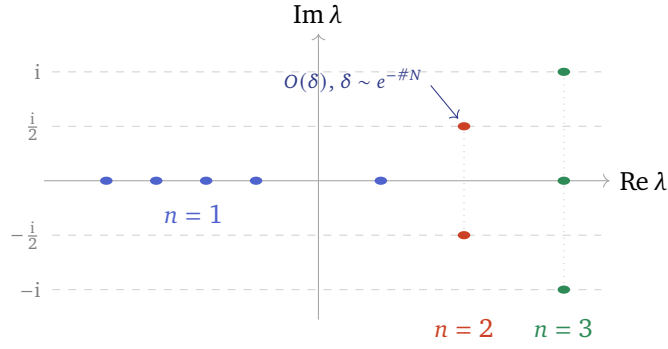


Figure 20.5: The string hypothesis: every root belongs to a string $\lambda^n + \frac{i}{2}(n+1-2a)$, $a = 1, \dots, n$, centred on the real axis and symmetric about it, up to deviations exponentially small in N — so a 2-string sits at $\pm \frac{i}{2}$ and a 3-string at $0, \pm i$. Real roots are 1-strings; an n -string is a bound state of n magnons. It is the *centres* λ^n that the thermodynamics treats as particles.

· *cited, and false in detail: Essler, Korepin and Schoutens (1992)*

The hypothesis is known to fail. There are “wide pairs” whose imaginary parts are not $\pm \frac{1}{2}$, and 2-strings that collapse onto the real axis as N grows; [Lecture 19](#) met the tail of the same phenomenon in the pair of real roots sharing a quantum number at $N = 22$ ([Remark 19.2](#)). What survives — and what the thermodynamics needs — is the *counting*: the string configurations reproduce the right number of states even where individual solutions are wrong. That is less than it sounds, and [§ 20.5](#) returns to it: a partition function weights states by *energy*, so equal cardinality is not equal free energy. This is an uncomfortable position to be in and it is the honest one.

Bethe–Takahashi

Put the string of [Definition 20.10](#) into the Bethe equations and multiply the n equations of its members together. The exponentially small deviations cancel between a diverging and a vanishing factor, and what is left is one equation per string *centre*:

$$N\theta_n(\lambda_\alpha^n) - \sum_{m,\beta} \Theta_{nm}(\lambda_\alpha^n - \lambda_\beta^m) = 2\pi I_\alpha^n, \quad \theta_n(\lambda) = 2 \arctan \frac{2\lambda}{n}, \quad (20.8)$$

with the scattering phase between an n -string and an m -string

$$\Theta_{nm} = \theta_{|n-m|} + 2\theta_{|n-m|+2} + \dots + 2\theta_{n+m-2} + \theta_{n+m}, \quad \theta_0 \equiv 0. \quad (20.9)$$

· *sketched: the cancellation of the deviations is the content, and it is cited (Takahashi 1971)*

For $n = m = 1$ this is $\Theta_{11} = \theta_2$ and (20.8) is [Lecture 19](#)’s (19.1). Two consequences follow as they did there, by reading off the values at $\pm\infty$: since $\Theta_{nm}(\pm\infty) = \pm\pi t_{nm}$ with $t_{nm} = 2 \min(n, m) - \delta_{nm}$, the number of vacancies for n -strings is

$$\#\{\text{vacancies for } n\text{-strings}\} = N - \sum_m t_{nm} M_m, \quad \rho_n + \rho_n^h = a_n - \sum_m A_{nm} * \rho_m, \quad A_{nm} = \frac{\Theta'_{nm}}{2\pi}, \quad (20.10)$$

with M_m the number of m -strings and $a_n = \theta'_n/2\pi$ as in [Theorem 19.3](#). The energy follows from the identity $2 \sum_{a=1}^n (4\lambda_a^2 + 1)^{-1} = \pi a_n(\lambda)$ for an exact string, together with [Lecture 15](#)’s $E = -\sum_j 2J/(4\lambda_j^2 + 1)$ (15.16): an n -string of centre λ carries energy $-\pi J a_n(\lambda)$ and S^z lowered by n . For $n = 1$ that is $\varepsilon_0 = -\pi J a_1$, the bare energy [§ 20.2](#) asked for.

The chain's TBA, and the Y-system

Repeat § 20.2 with one density per string length. Put the chain in a field, $H \mapsto H - h \sum_n S_n^z$. An n -string lowers S^z by n , so the Zeeman energy is $-hN/2 + h \sum_n nM_n$: the magnons' chemical potential is $-h$, and it is $+h$ that takes the place of the gas's $-\mu$. Nothing else changes except the bookkeeping, and the answer is an infinite coupled system: with $\eta_n = e^{\varepsilon_n/T} = \rho_n^h/\rho_n$,

$$\varepsilon_n = -\pi J a_n + hn + T \sum_m A_{nm} * \ln(1 + \eta_m^{-1}), \quad f = -\frac{h}{2} - T \sum_n \int a_n \ln(1 + \eta_n^{-1}) d\lambda. \quad (20.11)$$

· *sketched: this is Theorem 20.5's variation with one density per string length, not redone here*

The sign in front of the sum has flipped because $K_{XX}/2\pi = -a_2$ (Remark 19.4): at $n = m = 1$, (20.11) is (20.4). Infinitely many equations, each with infinitely many terms, is not progress. It becomes progress because the matrix $1 + A$ has an inverse that is almost diagonal: in Fourier variables $(1 + A)^{-1} = 1 - \hat{s}$ (shift in n) with $\hat{s}(\omega) = (2 \cosh(\omega/2))^{-1}$, i.e. with the kernel $s(\lambda) = (2 \cosh \pi \lambda)^{-1}$. Applying it collapses (20.11) to

Theorem 20.11 (Takahashi's TBA for the Heisenberg chain). *With $\eta_0 \equiv 0$ and $s(\lambda) = (2 \cosh \pi \lambda)^{-1}$,*

$$\ln \eta_n = -\delta_{n,1} \frac{\pi J}{T} s + s * \ln[(1 + \eta_{n-1})(1 + \eta_{n+1})], \quad f = -J \ln 2 - T \int s(\lambda) \ln(1 + \eta_1(\lambda)) d\lambda, \quad (20.12)$$

with $h \geq 0$ entering only through $\lim_{n \rightarrow \infty} (\ln \eta_n)/n = h/T$.

· *sketched: the Fourier inversion is checked in the script; the convergence of the truncation is numerical*

Three things fall out at once. The driving term now sits at $n = 1$ alone, and every other equation couples only to its two neighbours — a chain in the index n . The constant $-J \ln 2$ in f is Lecture 19's Hulthén ground-state energy (19.9), and it arrives here as $\int s \varepsilon_0 = -J \int s \pi a_1 = -J \ln 2$, not by assumption. And at $T \rightarrow \infty$ the driving vanishes and (20.12) has the constant solution $\eta_n = n(n+2)$, whence $f \rightarrow -T \int s \ln 4 = -T \ln 2$ — the free energy of N free spins, by a telescoping product.

Remark 20.12 (The Y-system). s is the inverse of $g \mapsto g(\lambda + \frac{i}{2}) + g(\lambda - \frac{i}{2})$, and $s(\lambda \pm \frac{i}{2}) = \mp (2i \sinh \pi \lambda)^{-1}$, whose sum vanishes identically. Those two facts use the same symbol for two prescriptions — approached from inside the strip the shifted sum of s is a nascent δ , which is exactly why the operator inverts $s*$ — so the driving term is not destroyed but *re-encoded*. For η_n analytic in $|\operatorname{Im} \lambda| \leq \frac{1}{2}$, applying the operator to (20.12) away from $\lambda = 0$ leaves

$$\eta_n(\lambda + \frac{i}{2}) \eta_n(\lambda - \frac{i}{2}) = (1 + \eta_{n-1}(\lambda))(1 + \eta_{n+1}(\lambda)). \quad (20.13)$$

The temperature, the coupling and the field have vanished: they survive in the analyticity and the asymptotics alone. *The Y-system is universal*, and it is where Lecture 18's Q reappears: fusing the auxiliary space gives a tower of transfer matrices whose eigenvalues obey the same hierarchy and are generated from a single Q (18.18). *The thermal η_n are not those ratios* — the same functional relation has many solutions, and the temperature is in the analyticity — which is the point the reading makes.

In practice one truncates at some $n_{\max}$ and, at zero field, imposes $\eta_n \rightarrow n(n+2)$ beyond it. At $T = J$ the truncation error falls below 10^{-5} by $n_{\max} = 20$, and the free energy so

computed agrees with exact diagonalisation of the finite chain — with a *finite* chain, where no string ever appears — to five decimals at $T = 2J$, J and $J/2$. The check script does exactly that comparison, and it is the strongest evidence *this lecture offers* that the string hypothesis, false in detail, counts correctly — not the strongest there is, since the same equations follow algebraically from the quantum transfer matrix with no string hypothesis at all — *cited: Essler et al. (2005)*. Note what it settles. That the string configurations preserve the *number* of states does not by itself give the free energy, since Z weights states by energy; it is the agreement of the *energies* that is the evidence.

20.6 The dictionary

Lecture 5 built action-angle variables and, with them, the measure $dI d\varphi$ on phase space: the angles are uniform on the torus, the actions label which torus, and Liouville's theorem says the measure is invariant. Classical statistical mechanics on an integrable system is the statistical mechanics of that measure — the angles carry no energy, so everything reduces to a distribution over the actions.

This lecture is the same sentence, quantised. The rapidity line has replaced the action space; a distribution over tori has become the pair (ρ, ρ_h) , or equivalently the single function ϑ ; and the invariant measure has become the *count* (20.1) of microstates carrying a given ϑ . Where the classical count is a volume, the quantum count is a binomial coefficient — the vacancies are discrete, and that discreteness is the entire difference between (20.1) and $\int dI$.

Dictionary

The invariant phase-space measure $dI d\varphi$ of action-angle variables (Lecture 5), and statistical mechanics as a distribution over the actions $\rightarrow$ the Yang–Yang entropy (20.1) of rapidity configurations, a count of which vacancies are filled (*type A*)

The classical partition function of an integrable system, $\int e^{-H(I)/T} dI d\varphi$: an integral over tori, the angles contributing only their volume $\rightarrow$ the Yang–Yang free energy, a *minimisation* over filling fractions ϑ (20.4), the interaction carried entirely by the kernel K (*type A*)

Reading

20.7 The field's factor, and where Q reappears

Two things § 20.5 asserted are worth doing properly: the field's normalisation, which no check of the equations can see, and the promise Lecture 18 made about the T–Q relation, which is worth keeping exactly as far as it is true.

The field. Everything in (20.12) except the boundary condition is blind to h : the operator that produced it annihilates the sequence h_n (check script). *So the normalisation of h survives in exactly one place, and it is the one place no check of the equations can see.* Set $J = 0$, where the answer is known. The recurrence has the constant solution $1 + \eta_n = (\sinh((n+1)x)/\sinh x)^2$, for which $(\ln \eta_n)/n \rightarrow 2x$, and (20.12) gives $f = -T \ln(2 \cosh x)$. A free spin in $H - h \sum S^z$ has $f = -T \ln(2 \cosh(h/2T))$, so $x = h/2T$ and the boundary condition is h/T . Takahashi's is $2h_T/T$ because his field term is $-2h_T \sum S^z$: *taking his boundary condition with our Hamiltonian multiplies the zero-field susceptibility by four.* As $x \rightarrow 0$ the hyperbolic solution becomes $\eta_n = n(n+2)$, which is why that is the zero-field truncation.

Where Q reappears, and where it does not. Fusing the auxiliary space of [Lecture 17](#) to spin $\frac{n}{2}$ gives a tower of transfer matrices T_n , and their eigenvalues obey a T -system: a relation of the shape

$$T_n^{[1]} T_n^{[-1]} = T_{n-1} T_{n+1} + f_n, \quad g^{[m]}(\lambda) = g\left(\lambda + \frac{im}{2}\right), \quad (20.14)$$

with f_n a known scalar built from the vacuum factors. Put $Y_n = T_{n-1} T_{n+1} / f_n$; then (20.14) at $n \mp 1$ gives $1 + Y_{n\mp 1} = T_{n\mp 1}^{[1]} T_{n\mp 1}^{[-1]} / f_{n\mp 1}$, and the Y_n satisfy (20.13) provided $f_n^{[1]} f_n^{[-1]} = f_{n-1} f_{n+1}$ — a condition on the scalar alone, which the standard f_n satisfies identically (check script). And every fused eigenvalue is generated from a single Baxter Q together with those vacuum factors: that is the sense in which [Lecture 18](#)'s T - Q relation (18.18) generalises · *cited: Kuniba, Nakanishi and Suzuki (2011) §§6, 8.*

Now the warning, which is the reason this is a reading and not a theorem. The Y_n so constructed are **not** the thermal η_n . They cannot be: an ordinary transfer matrix of a finite chain has no temperature in it. At $N = 2$ on the reference state the ratio $T_0 T_2 / f_1$ is a non-constant rational function of λ , negative on an interval, where the $J = h = 0$ thermal solution is the constant $\eta_1 = 3$. A functional equation does not determine its solution; the temperature enters through analyticity and asymptotics, and the object whose ratios are the thermal η_n is the quantum transfer matrix, a Trotter-limit construction this course does not build · *cited: Essler et al. (2005), §13.2 and appendix 13.A.* What [Lecture 18](#) promised is delivered here at the level of the hierarchy — the Y -system of (20.13) is the T - Q relation's, seen from the other side — and not at the level of the solutions.

Reading

20.8 Why it works, and what it does not prove

[Theorem 20.5](#) minimises a functional. What entitles anyone to believe that the minimum is the free energy?

The argument is a saddle point. Write the partition function as a sum over sets of quantum numbers, group the sets by their coarse-grained densities, and the sum becomes $Z = \sum_{\{\rho\}} e^{-Lg[\rho]/T}$ over the coarse-grained configurations, with g of (20.2): the number of terms in each group is e^{Ls} by [Proposition 20.1](#) and each contributes $e^{-L(e-\mu n)/T}$. As $L \rightarrow \infty$ the sum is dominated by its largest term and $-T \ln Z / L \rightarrow \min g$, provided the number of groups grows sub-exponentially — which is the usual requirement, and the usual thing that is waved through.

Yang and Yang proved what can be proved cleanly: that (20.4) has a unique solution for every $c > 0$, $T > 0$, that ε is analytic in μ and T and increasing in $|k|$, and that the resulting $P(\mu, T)$ is convex — so that the thermodynamics it defines is consistent · *cited: Yang and Yang (1969).* What their argument does *not* establish is the step before: that the set of Bethe states really is in bijection with the sets of quantum numbers, and that the coarse-graining loses nothing. For the gas, half of the first is a theorem: quantum numbers and solutions are in bijection ([Proposition 16.5](#)); that the states so built are *all* of them is still cited. For the chain even that much rests on the string hypothesis, which is false as a statement about individual solutions. The subject's position is that the counting is right even so. Part of the evidence is numerical — § 20.5's exact diagonalisation, at the three temperatures it reports, and no further, since a ring of fourteen sites is not the thermodynamic limit at low temperature. The rest is theoretical, and independent: the same equations come out of the quantum transfer matrix without any string hypothesis · *cited: Essler et al. (2005), §13.2 and appendix 13.A.*

There is a second, sharper reason to trust it. The same ϑ that minimises g is the one that appears in the dressing equation, and [Corollary 20.6](#) makes the dressed velocity a derivative of the free energy's own ingredients. Thermodynamics and dynamics are answered by one function. [Lecture 23](#) takes that seriously enough to let ϑ depend on position and time.

One more caution, which [Lecture 19](#) also raised. ε is not a *bare, state-independent* one-particle energy: it depends on T and μ , and only at $c = \infty$ does it reduce to $k^2 - \mu$. What it is is the *dressed* cost of a particle–hole excitation above the thermal state — moving a root from λ_h to λ_p costs $\varepsilon(\lambda_p) - \varepsilon(\lambda_h)$, backflow included · *cited: Korepin, Bogoliubov and Izergin §1.8* — which at $T = 0$ reduces to [Lecture 19](#)'s dressed energy. The trap is the next step: reading $e^{-\varepsilon/T}$ as a Boltzmann weight for a particle of energy ε gets the filling right for the wrong reason, and stops working the moment one asks for a correlation, where the shape of the state and not only its occupations is needed.

Tutorial

20.9 Solving the Yang–Yang equation

The recipe is [Lecture 19](#)'s Nyström method (19.21) wrapped in the iteration of [Proposition 20.9](#). Take an n -point Gauss–Legendre rule on $[-\Lambda, \Lambda]$, start from $\varepsilon = k^2 - \mu$, and iterate

$$\varepsilon_i \leftarrow k_i^2 - \mu - \frac{T}{2\pi} \sum_j w_j K(k_i - k_j) \ln(1 + e^{-\varepsilon_j/T}), \quad (20.15)$$

then read off P , ϑ , and — by one linear solve with the same matrix — ρ_t and every dressed quantity. Three practical points, all of which cost something before they were learnt.

The cutoff and the grid answer to different scales. Λ is set by where $\ln(1 + e^{-\varepsilon/T})$ dies — by $\sqrt{\mu + \#T}$, and read off the *interacting* ε , not the bare one. The number of nodes is set by the sharpest feature, and there are two: at low temperature the Fermi edge, of width $T/|\varepsilon'|$ in rapidity (*not* T/ν^{eff} , which is its width in dressed momentum), and at small c the kernel itself, a peak of width c . Large c is therefore the *easy* end — the kernel flattens towards $2/c$ — and small c is the hard one, where a rule that does not resolve the peak has a Nyström diagonal $w_i/\pi c$ that diverges instead of tending to a row integral of one. What is not set by c is Λ : taking $\Lambda \propto c$, as looked natural, put the nodes so far apart that the $c = 100$ pressure came out as 10^{-6} .

Convergence in n is not uniform in T . At $c = \mu = 1$ with the cutoff $\Lambda = \sqrt{\mu + 60T} + 12$ used throughout (so $\Lambda = 16.0$ at $T = 0.25$ and 13.3 at $T = 0.01$), an 800-point rule is converged to ten digits at $T = 0.25$, where 200 points give 3×10^{-5} ; at $T = 0.01$ the same 200 points are wrong in the *fourth* digit, for exactly the reason above. *Quote a cutoff with every such number* — at $\Lambda = 20$ the warm 200-point error is six times larger. As in [Lecture 19](#): run n against $2n$, and do it at the temperature you mean to use, not at the one where the code was written.

Check the thermodynamics, not the equation. The strongest test costs two extra solves: n against $\partial P/\partial \mu$ and s against $\partial P/\partial T$, with s computed from (20.1) and n from $\int \rho$. These agree to eight digits when everything is right and fail loudly when a factor of 2π is not. They also confirm the equation's weakest link — ρ_t comes from a linear solve that never appears in (20.4), so nothing else would catch an error in it.

A worked number. At $c = 1$ and the $\mu = 1.5554$ that [Lecture 19](#) gives for $q = 1.5$ (so $\gamma = 0.930$, $e = 0.604$), the sound velocity is $\nu^{\text{eff}} = 1.9086$ and the low-temperature pressure obeys $P(T) - P(0) = \pi T^2/6\nu^{\text{eff}}$ to four digits at $T = 0.02$: the Sommerfeld correction with

central charge one, which is the Luttinger-liquid statement of [Lecture 19](#)’s reading seen from the thermodynamic side.

Finally, the chain. The same iteration solves (20.12), but now s^* is a convolution best done by fast Fourier transform on a uniform grid, the index n must be truncated, and the initial guess $\eta_n = n(n+2)$ is the right one because it is the exact $T = \infty$ solution. Twenty string lengths and a grid of 2^{12} points on $|\lambda| \leq 40$ give the free energy to five decimals in about six seconds (thirty lengths cost three times that and buy one more decimal), which is enough to put it beside the exact diagonalisation of a 14-site ring and watch the two agree.

Summary.

- The entropy (20.1) is a count of which vacancies are filled: $s = \int [\rho_t \ln \rho_t - \rho \ln \rho - \rho_h \ln \rho_h]$ ([Proposition 20.1](#)), fermionic in form although the gas is bosonic ([Remark 20.3](#)).
- The Yang–Yang equation (20.4): minimising $e - \mu n - Ts$ with ρ_h constrained by (20.3) gives one nonlinear integral equation for the pseudo-energy, with $\vartheta = 1/(1 + e^{\varepsilon/T})$ and $\min g = -\frac{T}{2\pi} \int p'_0 \ln(1 + e^{-\varepsilon/T})$ ([Theorem 20.5](#)) — the weight matters; for the gas $p'_0 = 1$ and $-\min g$ is the pressure P . Varying ρ and ρ_h independently destroys the interaction ([Remark 20.4](#)).
- Dressing survives: $\varepsilon' = (\varepsilon'_0)^{\text{dr}}$ at every temperature, from the nonlinear equation and with no edge condition; and for the gas $v^{\text{eff}} = \varepsilon'/2\pi\rho_t$ is then defined ([Corollary 20.6](#), [Remark 20.7](#)).
- The checks: $c \rightarrow \infty$ gives free fermions in a box lengthened by $2/c$ per particle — a negative rod length, which is why the pressure exceeds the impenetrable one; $T \rightarrow 0$ at $\mu > 0$ gives [Lecture 19](#), with $\varepsilon(\pm q) = 0$ deduced rather than imposed ([Proposition 20.8](#)). The equation of state is measured (van Amerongen *et al.* 2008).
- Strings ([Definition 20.10](#)): complex roots organise into $\lambda^n + \frac{1}{2}(n+1-2a)$; the hypothesis is false in detail and counts correctly for the states, which is less than counting the free energy correctly. Bethe–Takahashi (20.8) and $N - \sum_m t_{nm} M_m$ vacancies (20.10).
- The chain’s TBA is infinite (20.11) and collapses to a nearest-neighbour chain in n (20.12) ([Theorem 20.11](#)), whose $T = 0$ constant is Hulthén’s $-J \ln 2$; the driving term is invisible to the Y-system (20.13) ([Remark 20.12](#)).

Tutorial. Exercise Sheet 20.

References

C. N. Yang and C. P. Yang, *J. Math. Phys.* **10** (1969) 1115 — the lecture’s §20.1–§20.4 in fourteen pages; M. Takahashi, *Thermodynamics of One-Dimensional Solvable Models*, Cambridge University Press (1999), chapter 7 (the repulsive gas) and chapter 8 (the XXX chain: §§8.1–8.2 the strings, §§8.3–8.4 the TBA equations and the free energy); chapters 3–5 are the ground state and the excitations. The original is M. Takahashi, *Prog. Theor. Phys.* **46** (1971) 401; V. E. Korepin, N. M. Bogoliubov and A. G. Izergin, *Quantum Inverse Scattering Method and Correlation Functions*, Cambridge University Press (1993), §I.6 (the Yang–Yang equation and its thermodynamics); F. Franchini, *An Introduction to Integrable Techniques for One-Dimensional Quantum Systems*, Springer (2017), §2.9 (“Thermodynamics of the model: the Yang–Yang equation”), whose §§2.9.1–2.9.3 are this lecture’s two limits and the weak-coupling one it does not do. On the failure of the string hypothesis: F. H. L. Essler,

V. E. Korepin and K. Schoutens, J. Phys. A **25** (1992) 4115. For the route that avoids it — the quantum transfer matrix and the fusion hierarchy — F. H. L. Essler, H. Frahm, F. Göhmann, A. Klümper and V. E. Korepin, *The One-Dimensional Hubbard Model*, Cambridge University Press (2005), §§12–13. On Y-systems: A. B. Zamolodchikov, Phys. Lett. B **253** (1991) 391, and the review A. Kuniba, T. Nakanishi and J. Suzuki, J. Phys. A **44** (2011) 103001. The experiment: A. H. van Amerongen, J. J. P. van Es, P. Wicke, K. V. Kheruntsyan and N. J. van Druten, Phys. Rev. Lett. **100** (2008) 090402.